

PARTH VINOD PATIL

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Summary

A Software Developer with 3+ years of industry experience and proven expertise in developing software for autonomous systems; proficient with machine learning, full-stack development, cloud computing, and algorithms, with a strong passion for learning.

Education

Purdue University	Master of Science in Electrical & Computer Engineering	[Jan' 24 - Present]
Indian Institute of Technology – Bombay (IIT-B)	B.Tech in Electrical Engineering with honors	[Aug' 17 - Jun' 21]

Publications

<i>ICSE</i>	Recommending Pre-Trained Models for IoT Devices PV Patil, W Jiang, H Peng, et al.	2025
	47th IEEE/ACM International Conference on Software Engineering - SERP4IoT Workshop	
<i>ICSE</i>	Enabling Unit Proofing for Software Implementation Verification PC Amusuo, PV Patil, et al.	2025
	47th IEEE/ACM International Conference on Software Engineering - NIER Track	

Professional Experience

Amazon Robotics	Manufacturing Test Engineer co-op	[Jan' 25 - Aug' 25]
	- Contributed to the development of a manufacturing test solution for a state-of-the-art end-of-arm tooling system. - Engineered a communication protocol between PLC hardware and a Python application to enable automated testing.	
Drivetrain Ai Technologies	Software Development Engineer 2	[Jan' 23 - Jan' 24]
	- Upgraded website-wide search with ElasticSearch, boosting hit rate by 3x and delivering search results within 200ms. - Built a ChatGPT integration for search, automating actions, and reducing metric creation time from 10 mins to a few seconds.	
Udaan	Robotics Engineer 2	[Jul' 21 – Jan' 23]
	- Led the development of autonomous swarm robots based Goods-To-Person system in a warehouse called 'Vahaan', capable of lifting racks of 500kgs (a high payload weight class) and navigating the layout avoiding human obstacles - Deployed multi-robot path planning algorithm based on D* & node reservation, yielding zero collision paths. - Architected the Fleet Management Service, responsible for optimal task allocation, enabling faster picking for items.	
Samsung Electronics	Research Intern	[May' 20 - Jul' 20]
	Collaborated on the development of Samsung SmartThings Hub 3 by migrating device health monitoring from the cloud to a Hub 3 based system with edge computing and using Kafka for syncing events thus saving \$23 million in cloud cost.	

Research & Projects

Formally verified controller synthesis from Natural Language	Research Guide: James C. Davis	[Aug' 24 – Present]
	Developing a method to synthesize controllers from natural language prompts verified with LTL specifications for correctness.	
Object Manipulation with four Legged Robots	Research Guide: A. Qureshi	[Jan' 24 – Jul' 24]
	Improvise a novel approach for using the leg of the quadruped as a manipulator using Unitree B1 inside Isaac Gym.	
Anomalous Human Activity Detection	Research Guide: R. Velmurugan	[Aug' 20 – Apr' 21]
	Engineered a novel framework for anomaly detection in CCTV videos, leveraging Prototypical CNN to analyze only 5 normal frames and identify anomalies such as robbery, accidents, over-speeding cars, etc in videos	
Autonomous Underwater Vehicle (AUV)	Software Subdivision Lead	[Sep' 17 – Jun' 21]
	- Fabricated 'Matsya', an autonomous underwater vehicle, with visual, acoustic, and depth sensors for realistic naval mission. - Implemented Extended Kalman filter-based sensor-fusion estimation that reduced pose estimation drifting by 90%. - Collaborated with 'Larsen & Toubro Defence' to design an underwater Remotely Operated Vehicle (ROV) deployable in seawater for scanning ship hulls & for surveillance in pots and ocean.	
Django Software Foundation	Google Summer of Code (GSoC)	[May' 19 - Aug' 19]
	- Optimized FormSet and similar classes to use declarative syntax, boosting user-friendliness and productivity of forms. - Created an 'edit only' mode in ModelFormSet to prevent SQL injection attacks, improving overall security of models.	

Skills and Experience

Languages	Python, C++, C#, Java, Kotlin, Bash, C, Groovy, Rust, Ruby, Go, Assembly, VHDL, ST
Frameworks	• Robotics : ROS, Gazebo, IsaacGym, Codesys • AI/ML : PyTorch, OpenCV, TensorFlow, Pandas
Backend	React, Angular, JavaScript/Typescript, Django, Azure, AWS, GCP, Node, Docker, Kubernetes